

High-Level Design (HLD) Document Template

Reference Template for Document Validation System

1. Introduction

1.1 Purpose

Describe the purpose of this High-Level Design (HLD) document. Mention the system name and explain what this document is intended to define.

1.2 Scope

Define the scope of the design. Explain which components and modules are covered by this document.

1.3 Design Objectives

- Create a scalable and maintainable architecture
- Ensure system reliability and performance
- Support future enhancements and modifications

1.4 Definitions, Acronyms, and Abbreviations

List important terms, acronyms, and abbreviations used in the document.

Term	Description
HLD	High-Level Design
API	Application Programming Interface
Module	A self-contained component of the system

1.5 References

List related documents such as the SRS, FRD, project guidelines, and technical standards.

2. System Architecture

2.1 Architecture Overview

Provide a high-level overview of the system architecture and its major components.

2.2 Architecture Diagram

Insert the system architecture diagram showing the main components and their interactions.

2.3 Technology Stack

Describe the technologies and frameworks used across the system layers.

- Frontend: web-based user interface
- Backend: REST API server
- AI Engine: document analysis service
- Database: relational database for storage

3. Component Design

3.1 Major Components

Describe the main components of the system and their responsibilities.

Component	Purpose	Technology
Web UI	Interface for document upload and viewing	Web framework
API Server	Business logic and request handling	REST API
Document Processor	Parse and extract text from documents	Extraction library
Validation Engine	AI-based document validation	AI service
Database	Persistent data storage	Relational DB

3.2 Component Interactions

Describe how the components communicate with each other to complete a validation request.

4. Data Design

4.1 Data Architecture

Describe the high-level data architecture and storage strategy.

4.2 Database Design

Provide an overview of the database and the main tables used by the system.

- Users - user credentials and profile information
- Documents - uploaded document metadata
- Templates - reference document templates
- Validation Results - validation scores and reports

4.3 Data Flow

Describe how data flows through the system from upload to validation result.

5. Interface Design

5.1 External Interfaces

Describe integration with external systems, APIs, or services.

- AI service for document analysis
- Email service for notifications
- Cloud storage for document archiving

5.2 API Overview

Provide a high-level overview of the key APIs. Detailed specifications are provided in the LLD.

6. Security and Performance

6.1 Security Design

- Authentication using secure tokens
- Role-based access control for restricted features
- Data encryption in transit and at rest
- Secure password storage using hashing

6.2 Performance and Scalability

- Horizontal scaling for API servers
- Database optimization with indexing and caching
- Asynchronous processing for long-running tasks

7. Deployment

7.1 Deployment Strategy

Describe the deployment approach, such as cloud, on-premise, or hybrid.

- Cloud-based deployment
- Containerization for consistent environments
- Automated build and deployment pipeline

7.2 Infrastructure Requirements

Provide an overview of the infrastructure needs. Detailed requirements are provided in the LLD.

8. Validation Rules

8.1 Required Sections

The uploaded HLD document should include the following main sections:

- Introduction
- System Architecture
- Component Design
- Data Design
- Interface Design
- Security and Performance
- Deployment
- Validation Rules
- Appendices

8.2 Validation Criteria

The document validation system should check whether required sections are available, whether section names are similar to the HLD template, whether sections follow the correct order, and whether the document belongs to the correct document type.

8.3 Scoring Criteria

Score Range	Validation Result
75-100	Accept
50-74	Needs Improvement
0-49	Reject

9. Appendices

9.1 Appendix A: Architecture Diagrams

Include detailed architecture and component diagrams.

9.2 Appendix B: Additional Information

Include any extra information related to the system design.